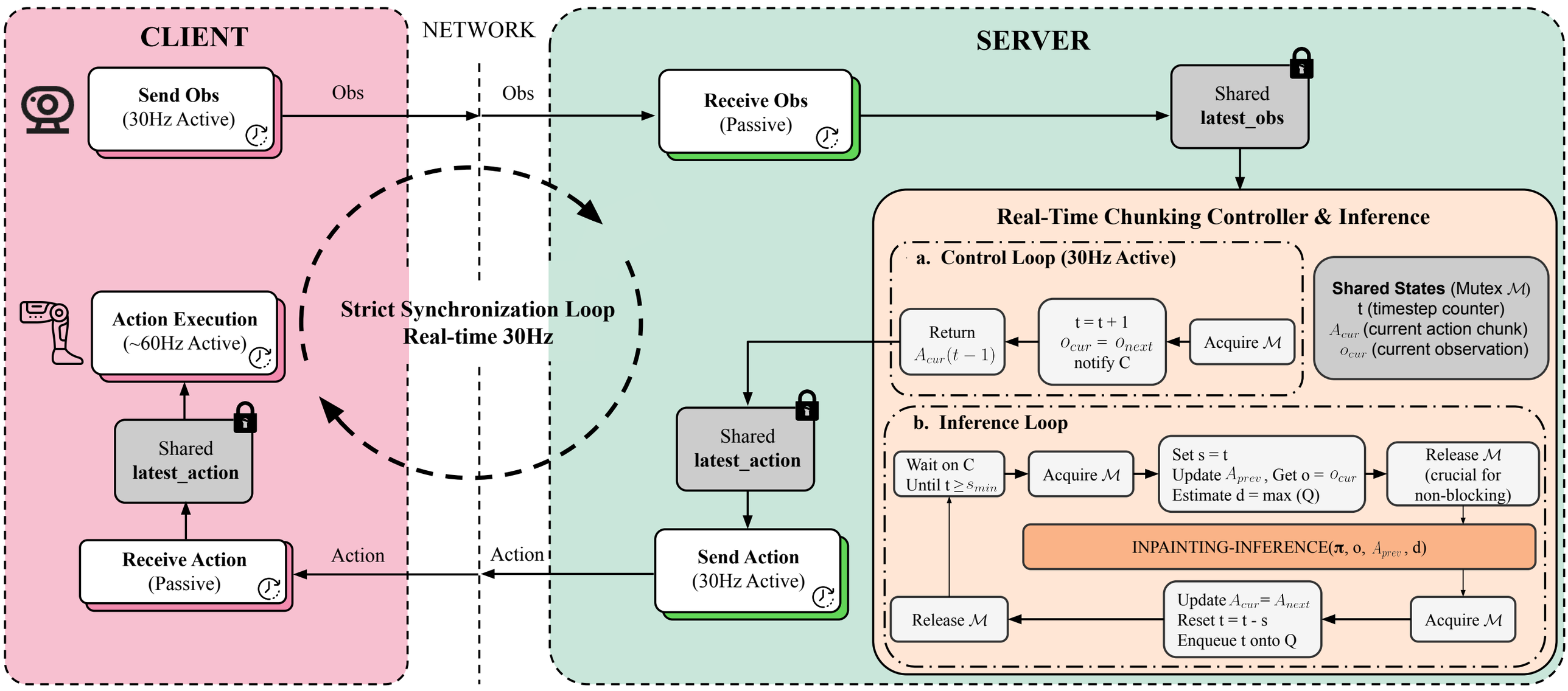
CLIENT
NETWORK
SERVER
Send Obs
(30Hz Active)
Obs
Obs
Receive Obs
(Passive)
Shared
latest_obs
Real-Time Chunking Controller & Inference
a. Control Loop (30Hz Active)
Return
$A_{cur}(t-1)$
t = t + 1
$o_{cur} = o_{next}$
notify C
Acquire $\mathcal{M}$
Shared States (Mutex $\mathcal{M}$)
t (timestep counter)
A_{cur} (current action chunk)
o_{cur} (current observation)
Strict Synchronization Loop
Real-time 30Hz
Action Execution
(~60Hz Active)
Shared
latest_action
Shared
latest_action
b. Inference Loop
Wait on C
Until t $\geq s_{min}$
Acquire $\mathcal{M}$
Set s = t
Update A_{prev}, Get o = o_{cur}
Estimate d = max (Q)
Release $\mathcal{M}$
(crucial for
non-blocking)
INPAINTING-INFERENCE(π, o, A_{prev}, d)
Release $\mathcal{M}$
Update $A_{cur} = A_{next}$
Reset t = t - s
Enqueue t onto Q
Acquire $\mathcal{M}$
Receive Action
(Passive)
Action
Action
Send Action
(30Hz Active)